

Reading Files

```
In [31]: import pandas as pd
```

```
In [32]: df = pd.read_csv(r"countries of the world.csv")
df
```

```
Out[32]:
```

	Country	Region
0	Afghanistan	ASIA (EX. NEAR EAST)
1	Albania	EASTERN EUROPE
2	Algeria	NORTHERN AFRICA
3	American Samoa	OCEANIA
...	...	...
223	Western Sahara	NORTHERN AFRICA
224	Yemen	NEAR EAST
225	Zambia	SUB-SAHARAN AFRICA
226	Zimbabwe	SUB-SAHARAN AFRICA

227 rows × 2 columns

```
In [33]: #df = pd.read_csv(r"C:\Users\alex\OneDrive\Documents\Pandas Tutorial\countries of
#df

df = pd.read_table(r"countries of the world.csv", sep = ',')
df
```

Out[33]:

	Country	Region
0	Afghanistan	ASIA (EX. NEAR EAST)
1	Albania	EASTERN EUROPE
2	Algeria	NORTHERN AFRICA
3	American Samoa	OCEANIA
...	...	...
223	Western Sahara	NORTHERN AFRICA
224	Yemen	NEAR EAST
225	Zambia	SUB-SAHARAN AFRICA
226	Zimbabwe	SUB-SAHARAN AFRICA

227 rows × 2 columns

```
In [34]: df = pd.read_json(r"json_sample.json")
df
```

Out[34]:

	12 Strong	A Fantastic Woman (Una Mujer Fantástica)	All The Money In The World	Bilal: A New Breed Of Hero	...	The Shape Of Water	Thriller: Billboard Outside Ebbin Missot
0	{'Genre': 'Action', 'Gross': '\$453,173', 'IMDB...'	{'popcornscore': 83, 'rating': 'R', 'tomatosco...	{'popcornscore': 71, 'rating': 'R', 'tomatosco...	{'popcornscore': 91, 'rating': 'PG13', 'tomato...	...	{'Genre': 'Adventure', 'Gross': '\$448,287', 'l...	{'popcornscore': 87, 'rating': 'tomatosco...

1 rows × 38 columns



```
In [35]: df2 = pd.read_excel(r"world_population_excel_workbook.xlsx", sheet_name = 'Sheet1')
df2
```

Out[35]:

	Rank	CCA3	Country	Capital
0	36	AFG	Afghanistan	Kabul
1	138	ALB	Albania	Tirana
2	34	DZA	Algeria	Algiers
3	213	ASM	American Samoa	Pago Pago
...	...	...	...	...
230	172	ESH	Western Sahara	El AaiËn
231	46	YEM	Yemen	Sanaa
232	63	ZMB	Zambia	Lusaka
233	74	ZWE	Zimbabwe	Harare

234 rows × 4 columns

```
In [36]: pd.set_option('display.max.rows', 8)
pd.set_option('display.max.columns', 8)
```

```
In [37]: df2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 234 entries, 0 to 233
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Rank        234 non-null    int64
1   CCA3        234 non-null    str
2   Country     234 non-null    str
3   Capital     234 non-null    str
dtypes: int64(1), str(3)
memory usage: 12.1 KB
```

```
In [38]: df2
```

Out[38]:

	Rank	CCA3	Country	Capital
0	36	AFG	Afghanistan	Kabul
1	138	ALB	Albania	Tirana
2	34	DZA	Algeria	Algiers
3	213	ASM	American Samoa	Pago Pago
...	...	...	...	...
230	172	ESH	Western Sahara	El AaiÃ°n
231	46	YEM	Yemen	Sanaa
232	63	ZMB	Zambia	Lusaka
233	74	ZWE	Zimbabwe	Harare

234 rows × 4 columns

In [43]: df2.shape

Out[43]: (234, 4)

In [44]: df2.tail(10)

Out[44]:

	Rank	CCA3	Country	Capital
224	43	UZB	Uzbekistan	Tashkent
225	181	VUT	Vanuatu	Port-Vila
226	234	VAT	Vatican City	Vatican City
227	51	VEN	Venezuela	Caracas
...	...	...	...	...
230	172	ESH	Western Sahara	El AaiÃ°n
231	46	YEM	Yemen	Sanaa
232	63	ZMB	Zambia	Lusaka
233	74	ZWE	Zimbabwe	Harare

10 rows × 4 columns

In [45]: df2.head(10)

Out[45]:

	Rank	CCA3	Country	Capital
0	36	AFG	Afghanistan	Kabul
1	138	ALB	Albania	Tirana
2	34	DZA	Algeria	Algiers
3	213	ASM	American Samoa	Pago Pago
...	...	...	...	...
6	224	AIA	Anguilla	The Valley
7	201	ATG	Antigua and Barbuda	Saint John's
8	33	ARG	Argentina	Buenos Aires
9	140	ARM	Armenia	Yerevan

10 rows × 4 columns

In [46]: df2['Rank']

Out[46]:

0	36
1	138
2	34
3	213
...	...
230	172
231	46
232	63
233	74

Name: Rank, Length: 234, dtype: int64

In [47]: df2.columns

Out[47]: Index(['Rank', 'CCA3', 'Country', 'Capital'], dtype='str')

In [60]: df2[df2['Country'] == 'Uzbekistan'] *# df2['Country'] is boolean indexing*

Out[60]:

	Rank	CCA3	Country	Capital
224	43	UZB	Uzbekistan	Tashkent

In [61]: df2.loc[df2['Country'] == 'Uzbekistan'] *# same as above*

Out[61]:

	Rank	CCA3	Country	Capital
224	43	UZB	Uzbekistan	Tashkent

loc - label based indexing,

iloc - integer based indexing

```
In [62]: df2.iloc[224]           # iloc displaying row at index 224
```

```
Out[62]: Rank           43  
CCA3           UZB  
Country    Uzbekistan  
Capital     Tashkent  
Name: 224, dtype: object
```

```
In [63]: # Filters for Uzbekistan and returns only the Capital and Rank columns  
df2.loc[df2['Country'] == 'Uzbekistan', ['Capital', 'Rank']]
```

```
Out[63]:
```

	Capital	Rank
224	Tashkent	43